AP CALCULUS AB

Unit 1 • Lesson 1.16 practice • Answer key

Lesson 1.16 • Working with the Intermediate Value Theorem

5 multiple-choice questions and 1 free-response question • Answer key with explanations

Directions

Questions 1 to 5 are multiple choice: choose the best of the four answer choices. Question 6 is free response: show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

Question 1

NO CALCULATOR

Which hypothesis is required to apply the Intermediate Value Theorem to f on $[a, b]$?

(A) f is continuous on $[a, b]$.
(B) f is increasing on $[a, b]$.
(C) $f(a)$ and $f(b)$ have the same sign.
(D) f is a polynomial.

Solution

Answer: (A). Continuity on the closed interval is the hypothesis of the IVT.

Question 2

NO CALCULATOR

f is continuous on $[0, 4]$ with $f(0) = 3$ and $f(4) = -1$. Which must be true?

(A) $f(2) = 1$
(B) $f(c) = 0$ for some c in $(0, 4)$
(C) f has exactly one zero in $(0, 4)$.
(D) $f(c) = 4$ for some c in $(0, 4)$

Solution

Answer: (B). 0 is between -1 and 3. The IVT guarantees at least one zero, not exactly one.

Question 3

NO CALCULATOR

For $f(x) = \frac{1}{x}$, $f(-1) = -1$ and $f(1) = 1$, yet $f(x) \neq 0$ for every x. Why does this not contradict the IVT?

(A) 0 is not between -1 and 1.
(B) The IVT only applies to polynomials.
(C) f is not continuous on $[-1, 1]$.
(D) f is decreasing.

Solution

Answer: (C). f is undefined at $x = 0$, so the hypothesis of continuity fails.

Question 4

NO CALCULATOR

g is continuous with $g(0) = 2$, $g(1) = -1$, $g(2) = 1$, $g(3) = 4$. What is the least number of zeros g must have on $[0, 3]$?

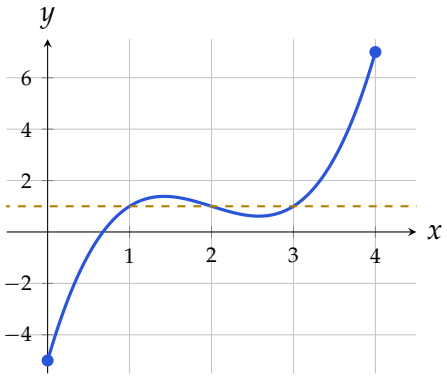
(A) 0
(B) 1
(C) 3
(D) 2

Solution
Answer: (D). Sign changes on $[0,1]$ and on $[1,2]$ give at least two zeros.

Question 5 (graph)

NO CALCULATOR

The graph of the continuous function $f(x) = (x-1)(x-2)(x-3)+1$ on $[0,4]$ is shown with the line $y = 1$ (dashed). How many values of c in $[0,4]$ satisfy $f(c) = 1$?



(A) Exactly one, as guaranteed by the IVT
(B) Exactly two
(C) Three
(D) None

Solution
Answer: (C). The IVT guarantees at least one value, since $f(0) = -5 < 1 < 7 = f(4)$. The graph shows that $f(c) = 1$ at $c = 1, 2$ and 3.

Question 6 (Free response)

CALCULATOR

Let $f(x) = x^3 - 2x - 2$.

(a) Use the IVT to show that f has a zero in $[1,2]$.
(b) Show that $f(c) = 10$ for some c in $[2,3]$.
(c) Use a calculator to approximate the zero of f in $[1,2]$ to three decimal places.
(d) Explain why the IVT cannot be applied to $g(x) = \dfrac{1}{x-1.5}$ on $[1,2]$.

Solution
(a) f is a polynomial, so it is continuous on $[1,2]$. $f(1) = -3 < 0$ and $f(2) = 2 > 0$. Since 0 is between -3 and 2, there is a c in $[1,2]$ with $f(c) = 0$.
(b) f is continuous on $[2,3]$, $f(2) = 2$ and $f(3) = 19$. Since $2 < 10 < 19$, the IVT gives a c in $[2,3]$ with $f(c) = 10$.
(c) $x \approx 1.769$.
(d) g is not continuous on $[1,2]$: it is undefined at $x = 1.5$.